

Support-ticket MCP server

Answers questions about 40,047 support tickets, says which ones the data cannot answer, and finds that half of the archive's first replies wasted a round trip.

A. DESIGN

Four MCP tools over stdio and no LLM in the server: the host supplies the model, the server supplies grounded facts and refusals. Corpus: 40,047 bilingual (en/de) tickets from Tobi-Bueck/customer-support-tickets, customer message, first reply, queue label. It is synthetic, built for classifier training.

- `get_schema`: columns, their values, where the rows came from (61,765 downloaded, 40,047 served), and an explicit `cannot_answer` list (no timestamps, customer ID, assignee or money), so the host refuses instead of confabulating.
- `run_sql`: one read-only SELECT against DuckDB.
- `find_similar_tickets`: multilingual-e5-small, mean-centred (unrelated pairs score ~0, not ~0.8); exact search over 31,625 train-split vectors. Below a 0.47 floor: `has_precedent`: false. Every hit carries `reply_state`.
- `suggest_routing`: k=1 nearest-neighbour vote (k swept on val; larger k floods small queues), with two confidence figures, top-1 similarity and top-3 agreement, fitted on validation and checked on test.

Fetch, eight stages, DuckDB and index: ~80s, deterministic, offline. Every threshold and metric the tools state is rendered from the build's files. Nothing here governs whether the host calls a tool, only what it gets back.

B. WHY

Support over Churn because Churn is one numeric table and SQL over it is solved; tickets are text plus fields, so retrieval fits the text, SQL fits the fields, and the host chooses. Three things deliberately avoided:

- **An LLM server-side.** The host already has one. A second would double the cost per call and make every tool non-deterministic

D. RISKS

An agent drafts a reply with no real precedent	Floor 0.47, fitted on validation: accepts 97.8% of real tickets, rejects 20 of 20 off-topic queries. Below it, no draft material. Typed questions score ~0.18 lower than tickets whatever their topic, partly because they are shorter, so for them the floor is advisory.
A confidence figure is over-trusted	Both fitted on validation and checked on test: in the upper bands similarity holds within 0.006 while agreement runs 0.034 high, so the tool calls that one a ceiling. When they differ by >0.15 the response names the lower as the conservative read.
Source language label wrong on 10% of rows	4,204 tickets labelled de carry English text. <code>language</code> is re-detected from the text; the original stays as <code>language_label</code> so the defect is queryable.
Silently wrong tag aggregates	Positional <code>tag_1..8</code> make <code>GROUP BY tag_1</code> run clean and wrong. Not served; <code>ticket_tags</code> holds one row per (ticket, tag). The wrong query is unwritable, not discouraged.
SQL reads files or mutates	Read-only, external access off, config locked, one SELECT via the parser, a cursor and limits per call. Capability removed, not names blacklisted. Residual: metadata functions reveal the database path.
Memorised test set inflates scores	8,357 tickets appear in both multi-language files; split first and ~28% of test has a twin in training, so dedup precedes the split and the index holds train only. Beyond exact copies, 55% of test tickets have a paraphrase in the index; on the rest, macro-F1 is 0.414. Routing is partly lookup.
German quietly worse than English	0.782 en / 0.542 de macro-F1. Coverage, not the encoder: retrieval is monolingual (96% of German queries' nearest neighbour is German) and the German slice of the index is 2.3x smaller; English cut to that size scores 0.545. Fix: more German tickets, not a bigger model.

E. BUSINESS IMPACT

Value is elapsed time, not agent minutes: a ticket settled in one exchange costs hours, one that needs three costs days, and the first reply decides which. "Send the system logs and your account number" can be acted on. "Please provide details" cannot.

49.8% of first replies in this archive are the second kind: they neither resolve the ticket nor name anything fetchable. (Haiku 4.5 over 39,739 answers, \$9.53; on the 40 I read by hand, model and reader both count 22 dead ends.) The archive is synthetic, so the rate describes the generator. What transfers is the measurement: the same labeller over CHEQ's real first replies gives the true rate

and unevaluable. The one LLM pass, labelling 39,739 first replies as dead end or not (E), ran once; its output is committed.

- **A fine-tuned classifier.** It would likely score higher, but returns a softmax with no evidence. kNN returns the neighbours, so a user can check the answer, and neighbour agreement becomes a second confidence signal.
- **A reranker.** The bias that suggested one was a bug: `subject + body` over-ranked subject-less tickets 2.6x; embedding the body alone removed it.

C. PRODUCTION

The four tool contracts stay; everything beneath them changes.

- **Source.** CSVs → DuckDB becomes Zendesk/Jira → Snowflake. The index rebuilds on an Airflow schedule and the corpus-relative floor recalibrates with it.
- **Transport.** stdio → streamable HTTP: same tool code, second endpoint, and where a `draft_reply` tool would live, since a webhook has no host model.
- **Access control.** None today. In production, SQL is covered by Snowflake row-access policies if the HTTP endpoint opens the session with the caller's own SSO token. The vector index is a copy of rows outside that boundary, so re-gate it: resolve the caller's scope, mask the index, then take top-k. Post-filtering leaks existence and guts top-k for narrow scopes.
- **Recency.** Weight the ranking by age, do not exclude: a three-year-old precedent may describe a dead product.
- **Scale.** Brute force measures 0.4 ms at 32k vectors and stays under 10 ms to about a million; an approximate index adds an error inseparable from the embedding's own, so not before.

in a day; `reply_state` on every precedent lets a draft copy the replies that ended the exchange, not the ones that extended it.

At CHEQ, two uses of one pattern. Support agents draft first replies from precedent over CHEQ's own archive (C, Source), with the dead-end rate before and after as the first business number. And the Data & AI team puts internal Snowflake data behind Claude for stakeholders (a CSM asking why an account's invalid-traffic share jumped) with this server as the template: refuse what the data can't answer, fit every threshold on held-out data, publish the confidence next to the label, so stakeholders trust numbers the data team did not hand-check.

Runs, evaluates and tests offline. Routing 0.707 macro-F1 over 10 queues against 0.045 for always answering the largest queue · `make build` 80s · `make eval` 30s · `make test` 2s · every evaluation number here is generated into `reports/`.